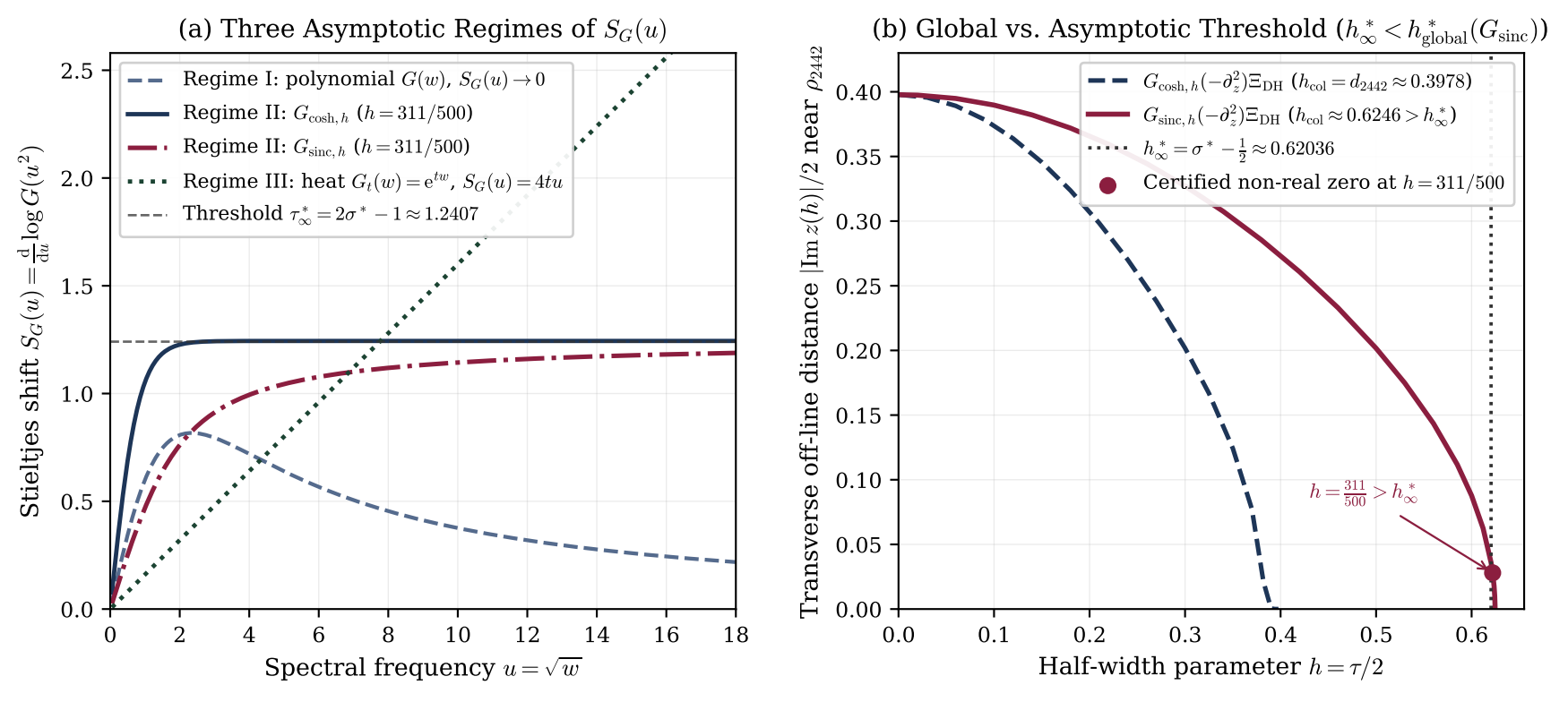

(a) Three Asymptotic Regimes of $S_G(u)$
Regime I: polynomial $G(w)$, $S_G(u) \to 0$
Regime II: $G_{\cosh,h}$ $(h = 311/500)$
Regime II: $G_{\mathrm{sinc},h}$ $(h = 311/500)$
Regime III: heat $G_t(w) = e^{tw}$, $S_G(u) = 4tu$
Threshold $\tau_\infty^* = 2\sigma^* - 1 \approx 1.2407$
Stieltjes shift $S_G(u) = \frac{d}{du} \log G(u^2)$
Spectral frequency $u = \sqrt{w}$
0.0
0.5
1.0
1.5
2.0
2.5
0
2
4
6
8
10
12
14
16
18
(b) Global vs. Asymptotic Threshold ($h_\infty^* < h_{\mathrm{global}}^*(G_{\mathrm{sinc}})$)
$G_{\cosh,h}(-\partial_z^2)\Xi_{\mathrm{DH}}$ $(h_{\mathrm{col}} = d_{2442} \approx 0.3978)$
$G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}$ $(h_{\mathrm{col}} \approx 0.6246 > h_\infty^*)$
$h_\infty^* = \sigma^* - \frac{1}{2} \approx 0.62036$
Certified non-real zero at $h = 311/500$
$h = \frac{311}{500} > h_\infty^*$
Transverse off-line distance $|\mathrm{Im}\, z(h)|/2$ near ρ_{2442}
Half-width parameter $h = \tau/2$
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.0
0.1
0.2
0.3
0.4
0.5
0.6